

Comments on the stabilizability of the angular velocity of a rigid body

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Abstract: This paper discusses the stabilizability by smooth feedback of the angular velocity of a rigid body. First we show how to *asymptotically* stabilize the system with a single control. Then it is shown that a single control aligned with a principal axis *cannot* asymptotically stabilize the system. Finally we present some new results on stabilization by means of scalar feedback control.

Keywords: Rigid body, Angular velocity, Feedback, Stabilization.

1. Introduction and motivation

In this paper we are concerned with the stabilization by smooth feedback of the angular velocity of a rigid body.

Consider a rigid body in an inertial reference frame. Let $\omega_1, \omega_2, \omega_3$ be the angular velocity components with respect to a body-fixed reference frame with origin at the center of gravity and consisting of the principal axes. Let I_1, I_2, I_3 denote the principal moments of inertia (positive real numbers). The Euler equations for the rigid body are

$$I\dot{\omega} = S(\omega)I\omega + Bu \quad (1)$$

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with $I = \text{diag}(I_1, I_2, I_3)$ the inertia matrix,

$$\omega \in \mathbb{R}^3, \quad B \in \mathbb{R}^{3 \times 3},$$

$$S(\omega) = \begin{pmatrix} 0 & \omega_3 & -\omega_2 \\ -\omega_3 & 0 & \omega_1 \\ \omega_2 & -\omega_1 & 0 \end{pmatrix},$$

$$u^T = (u_1, u_2, u_3) \in \mathbb{R}^3.$$

For later reference:

$$I_{23} = (I_2 - I_3)/I_1,$$

$$I_{31} = (I_3 - I_1)/I_2, \quad I_{12} = (I_1 - I_2)/I_3.$$

Notice that each u_i represents an external torque aligned with an axis with components given by the i -th column of B . There is obviously no loss of generality in assuming that there are at most three control axes.

The Euler angular velocity equations (1) have been an ever returning set of equations in the control theory literature. We will briefly discuss some well-known results related to the stabilizability aspects of the equations. Connections with the concept of controllability are left aside.

It has been shown in [1] that the null solution of Euler's angular velocity equations can be made *asymptotically stable* by means of *two* torques (each applied along a principal axis). This result has been rederived in [2] as an application of stabilization using center manifold theory. In addition, in [2], a new class of asymptotically stabilizing feedbacks has been presented. The problem whether or not asymptotic stabilizability can be achieved if only one control torque is available was left unanswered. Section 2 is dedicated to this question. It may come as a surprise that there exists a *scalar linear asymptotically stabilizing* feedback to Euler's angular velocity equations. This is the main contribution of the paper. It follows as a nice application of LaSalle's theorem. To our knowledge, this has not been reported before in the control literature.

Section 3 links Sections 2 and 3. First we state an important remark on the impossibility of asymptotically stabilizing the system by smooth feedback if there is only one control torque aligned with a principal axis. This will lead us to Section 4, where *stabilizability* aspects as opposed to *asymptotic stabilizability* aspects are investigated. Previous work has been done on stabilizability of Euler's angular velocity equations in [3] where stabilizing control laws which are *robust* with respect to small perturbations of the control laws are considered.

This control robustness aspect of the feedback law is further developed. We also take the opportunity to point out a 'caveat' in the main theorem in [3] on stabilizability of Euler's equations.

2. Asymptotic stabilization

2.1. Main Theorem

Consider as a feedback for (1)

$$u = -B^T \omega;$$

then the closed-loop system is of the form

$$\dot{\omega} = F(\omega) = I^{-1}S(\omega)I\omega - I^{-1}BB^T\omega. \quad (2)$$

The kinetic energy of the rigid body $V(\omega) = \frac{1}{2}\omega^T I \omega$ serves as a valid Ljapunov function. Indeed V is positive definite and zero at the origin. Its derivative along solutions of (2) is

$$\dot{V} = \omega^T I F(\omega) = \omega^T S(\omega)I\omega - \omega^T B B^T \omega$$

or

$$\dot{V} = -\|B^T \omega\|^2 \leq 0.$$

From Ljapunov's second theorem [4] it follows that $u = -B^T \omega$ is a *stabilizing* feedback. The following theorem is concerned with *asymptotic stabilizability*.

Main Theorem. *The feedback law $u = -B^T \omega$ makes the system (1) globally asymptotically stable at the origin if and only if each row of the control matrix B contains at least one nonzero element (this condition will be labeled (C)).*

Proof. By the expression of $\dot{V}$ above and by LaSalle's invariance principle, any trajectory of (2) converges to the largest invariant subset M of

$L = \{\omega: \dot{V}(\omega) = 0\}$. Notice that $L = \text{Ker } B^T$.

Our objective therefore is to show that $M = \{0\}$ is equivalent with condition (C).

Necessity. First we will show that (C) is equivalent with the fact that the controlled equation has no other equilibrium point than the origin. This implies that (C) is necessary for the asymptotically stabilizing property of $u = -B^T \omega$.

The equilibrium points ω_e of (2) are given by

$$I^{-1}S(\omega)I\omega - I^{-1}BB^T\omega = 0. \quad (3)$$

Since ω_e satisfies $\dot{V} = 0$ or $B^T \omega = 0$, (3) is equivalent with

$$S(\omega)I\omega = 0, \quad (4a)$$

$$B^T \omega = 0. \quad (4b)$$

A simple calculation shows that all ω_e satisfying (4) are of the form

$$\omega_e^T = (*, 0, 0)$$

or

$$\omega_e^T = (0, *, 0)$$

or

$$\omega_e^T = (0, 0, *)$$

with $(*) \in \mathbb{R}$. It is then immediate that the set of equations (4) has the origin as the only solution iff condition (C) is imposed. This proves necessity.

Sufficiency. We will show that (C) implies that $M = \{0\}$, by considering three cases according to the rank of the control matrix B .

(i) rank $B = 3$.

In this case $\{\omega: \dot{V} = 0\} = \text{Ker } B^T = \{0\}$, implying global asymptotic stability.

(ii) rank $B = 2$.

Then $L = \{\omega: \dot{V} = 0\} = \text{Ker } B^T$ is a line through the origin. For any trajectory $\omega(\cdot)$ belonging to L ,

$$\frac{d}{dt} V(\omega(t)) = 0, \quad t \geq 0.$$

Thus

$$V(\omega(t)) = c (\geq 0), \quad t \geq 0.$$

This trajectory $\omega(\cdot)$ then belongs to the ellipsoid $V_c = \{\omega: V(\omega) = c\}$, if $c > 0$, and to the line L through the origin and is therefore an equilibrium point. As shown above, condition (C) says that the

origin is the only equilibrium point (i.e. $c = 0$). This establishes global asymptotic stability for case (ii).

(iii) rank $B = 1$.

Assume without loss of generality that

$$\text{range } B = \text{span}\{b_1\},$$

with b_1 the first column of B ($b_1^T = (b_{11}, b_{21}, b_{31})$). By condition (C) all elements of b_1 are nonzero. Then

$$\begin{aligned} L &= \{\omega: \dot{V} = 0\} \\ &= \{\omega: b_{11}\omega_1 + b_{21}\omega_2 + b_{31}\omega_3 = 0\}. \end{aligned}$$

In order to characterize the largest invariant subset M of the plane L we notice that for each trajectory $\omega(\cdot)$ in L , $b_1^T\omega(t) = 0$ and therefore $b_1^T\dot{\omega}(t) = 0$. As a consequence any trajectory $\omega(t) \in L$ belongs to

$$\begin{aligned} \{\omega: b_1^T\omega = 0 \text{ and } b_1^TF(\omega) = 0\} \\ = \{\omega: b_1^TI^{-1}S(\omega)I\omega = 0\}. \end{aligned}$$

Thus M belongs to the intersection of the plane

$$L = \{\omega: b_{11}\omega_1 + b_{21}\omega_2 + b_{31}\omega_3 = 0\}$$

and the cone

$$\begin{aligned} C = \{\omega: b_{11}I_{23}\omega_2\omega_3 + b_{21}I_{31}\omega_3\omega_1 \\ + b_{31}I_{12}\omega_1\omega_2 = 0\}. \end{aligned}$$

Then $L \cap C$ consists of a finite number of lines through the origin. As before (see case (ii)) any trajectory in L also belongs to an ellipsoid $E = \{\omega: V(\omega) = c\}$ for some $c \geq 0$. Then $M \subset L \cap C \cap E$ and therefore M consists of isolated points different from the origin, if $c > 0$. But this is ruled out by condition (C) as shown before. Therefore $c = 0$ and the origin is a globally attracting equilibrium. This ends the sufficiency part.

2.2. Remarks

2.2.1. The main theorem remains valid for a smooth feedback $u = -DB^T\omega$, $\omega \in \mathbb{R}^3$, with D any positive definite matrix. This introduces some kind of robustness in the feedback law.

2.2.2. In the special case of *scalar* control, the theorem above states that the angular velocity of a rigid body can be asymptotically stabilized by $u = -b^T\omega$ if and only if all elements of the col-

umn vector b are nonzero. This means that the control torque is aligned with an axis having a nonzero component along all principal axes.

2.2.3. Obviously condition (C) of the theorem is *sufficient* for asymptotic stabilizability as such by means of feedback control. The condition is *necessary* for asymptotic stabilizability *by means of the specific control strategy* $u = -B^T\omega$. The condition is *not necessary* when *multi-input general feedback controls* are allowed. Indeed consider

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

corresponding to the case of two torques applied along principal axes. This system is asymptotically feedback stabilizable [1,2]. Whether the condition is necessary in the *scalar input case when general feedback laws are allowed* is an open question (see also Section 3).

3. Entr'acte

In the remaining part of the paper we will restrict ourselves to the angular velocity equations driven by *one* control torque:

$$I\dot{\omega} = S(\omega)I\omega + bu, \quad \omega \in \mathbb{R}^3, u \in \mathbb{R}, \quad (5)$$

or in coordinates

$$\begin{aligned} \dot{\omega}_1 &= I_{23}\omega_2\omega_3 + c_1u, \\ \dot{\omega}_2 &= I_{31}\omega_3\omega_1 + c_2u, \\ \dot{\omega}_3 &= I_{12}\omega_1\omega_2 + c_3u, \end{aligned} \quad (6)$$

with

$$\begin{aligned} b &= [b_1, b_2, b_3]^T, \\ c &= [c_1, c_2, c_3]^T, \quad c = I^{-1}b. \end{aligned}$$

If all $c_i \neq 0$, then by Section 2 there exists an asymptotically stabilizing feedback. What if some of the c_i are equal to zero? Assume $c_3 \neq 0$, $c_1 = c_2 = 0$. This means that the *control axis is aligned with a principal axis*. Then the system (6) is *not asymptotically stabilizable*. The proof follows by observing that

$$V \equiv I_{31}\omega_1^2 - I_{23}\omega_2^2$$

is a first integral of (6). This means that the level

surfaces $V_c = \{\omega: V(\omega) = c\}$ (i.e. elliptic or hyperbolic cylinders, depending on the sign of $I_{31}I_{23}$) are invariant manifolds, precluding asymptotic stabilizability.

The question of asymptotic stabilizability with one torque perpendicular to a principal axis (e.g. $c_1 \neq 0$, $c_2 \neq 0$, $c_3 = 0$) remains open.

4. Stabilizability

Consider again the system (6) with scalar control. Let the control torque be aligned with a principal axis ($c_1 = c_2 = 0$, $c_3 \neq 0$). Since asymptotic stabilizability is impossible for this system we relax our goal to an implementation of stabilizing feedback.

First notice that there is *obvious stabilizing control* for (6). Take $u \equiv 0$. Then the system is stable, since $V(\omega) = \frac{1}{2}\omega^T I \omega$ is a positive definite function with

$$\frac{d}{dt} V(\omega(t)) \equiv 0.$$

However, a small deviation from the zero input makes the system *unstable*. Indeed, let

$$u = \varepsilon \omega_3$$

(sign $\varepsilon = \text{sign } c_3$); then

$$\frac{d}{dt} V(\omega(t)) = I_3 \omega_3 c_3 u = \varepsilon c_3 I_3 \omega_3^2 \geq 0.$$

We must therefore look for *robust stabilizing control feedbacks*: i.e. such that addition of *small feedback control terms* does not corrupt its stabilizing potential.

A basic theorem [3] related to this question is as follows:

The null solution of Euler's angular velocity equations can be robustly stabilized with a smooth feedback control torque aligned with a principal axis if the moment of inertia along this axis is either *larger* or *smaller* than the remaining moments of inertia.

In the remainder we will be basically concerned with the case not mentioned in the theorem above:

$$I\dot{\omega} = S(\omega)I\omega + bu,$$

$$\omega \in \mathbb{R}^3, u \in \mathbb{R}^1, b = (0, b, 0),$$

or in coordinates

$$\begin{aligned} \dot{\omega}_1 &= I_{23}\omega_2\omega_3, \\ \dot{\omega}_2 &= I_{31}\omega_3\omega_1 + cu, \\ \dot{\omega}_3 &= I_{12}\omega_1\omega_2, \end{aligned} \tag{7}$$

with $c = b/I_2$ and with I_2 lying between I_1 and I_3 . Without loss of generality assume $I_1 > I_2 > I_3 > 0$. Then $I_{23} > 0$, $I_{12} > 0$, $I_{31} < 0$.

Let $u = -c\omega_2$. Consider

$$V(\omega) = \frac{1}{2}\omega^T I \omega, \quad \omega \in \mathbb{R}^3;$$

then

$$\frac{d}{dt} V(\omega(t)) = -cI_2\omega_2^2.$$

By LaSalle's theorem, each trajectory converges towards the largest invariant subset M of the plane $\omega_2 = 0$ as $t \rightarrow \infty$. A simple argument shows that

$$\begin{aligned} M &= \{\omega: \omega = (\omega_1, 0, 0) \\ &\quad \text{or } \omega = (0, 0, \omega_3); \omega_1, \omega_3 \in \mathbb{R}\} \end{aligned}$$

with each point in M an equilibrium point. Notice that for (7), the sets defined by

$$\{\omega: I_{12}\omega_1^2 - I_{23}\omega_3^2 = \text{constant}\}$$

are *invariant* hyperbolic cylinders and that *each* patch of such an invariant cylinder is intersected by M in exactly one equilibrium point. Thus each trajectory stays on an invariant patch of a cylinder and is attracted by the unique equilibrium point on that patch. This implies stability. In fact we have asymptotic stability *when restricting to the invariant manifolds*. In [3] we stated that the system (7) was not stabilizable. This is not true as shown above. However the choice of the stabilizing control $u = -c\omega_2$ is critical. Small deviations from this feedback (i.e. addition of small terms $k\omega_1$ and/or $l\omega_3$) annihilates its stabilizing potential, making the system *unstable*. The reader is referred to Remark 2 of this section for more information.

Remark 1. The control $u = -c\omega_2$ (or more generally $u = -\text{sign } c \cdot \alpha\omega_2$, with $\alpha > 0$) is in fact stabilizing irrespective of the relations between the moments of inertia. This follows from $V = \frac{1}{2}\omega^T I \omega$, with $\dot{V} = -c^2 I_2 \omega_2^2 \leq 0$. A major point to notice is that this control is robust if I_2 is larger or smaller

than I_1 and I_3 [3]. The control is non-robust if I_2 lies between I_1 and I_3 (see Remark 2).

Remark 2. The stabilizing aspects of $u = -c\omega_2$ for the system (7) with $I_{23} > 0$, $I_{12} > 0$, $I_{31} < 0$ can also be derived by the center manifold approach. We will not go through the computational details (although straightforward) but give a qualitative description of what is going on. By center manifold theory, the stability properties of the controlled system are adequately described by the stability properties of a (reduced) system in the plane (ω_1, ω_3) . This reduced system has level curves described by the *hyperbolas* $I_{23}\omega_3^2 - I_{12}\omega_1^2 = \text{constant}$ with asymptotes given by $I_{23}\omega_3^2 - I_{12}\omega_1^2 = 0$. In [3] we concluded that the reduced system and therefore the original system was unstable. This conclusion is erroneous as shown above and as can be seen from the center manifold approach. Indeed the control $u = -c\omega_2$ has the property that *each* hyperbolic invariant level curve of the reduced system is intersected with a curve of equilibrium points: it consists of an equilibrium point and two trajectories *converging* to that point. Thus each trajectory is kept from running to infinity as $t \rightarrow \infty$, by the introduction of an attracting equilibrium point at each level curve.

As for an explanation of the non-robustness property of the control $u = -c\omega_3$ mentioned above, consider

$$u = -c\omega_3 + k\omega_1 + l\omega_2,$$

with k and/or l nonzero; then a straightforward

calculation shows that there is a set of level curves which does not contain an equilibrium point. Each trajectory that starts on such a level curve runs off to infinity implying instability of the reduced system and therefore of the original system.

Remark 3. It is stressed that the Ljapunov approach does not lend itself easily to studying other (i.e. more general) feedback laws. We refer the reader to [3] for more information. Also in handling the robustness properties of a general class of stabilizing control laws the center manifold approach [3] proves to be superior.

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